supply-based CII which uses GT as the capacity is referred to as cgDIST.

3 Application

3.1 For all ships to which regulation 28 of M ARPOL Annex VI applies, the operational carbon intensity indicators defined in section 4 should be applied.

3.2 The operational carbon intensity indicators defined in section 5 are encouraged to be

---CHUNK---

SOURCE: MEPC.336(76).pdf
3.1 For all ships to which regulation 28 of M ARPOL Annex VI applies, the operational carbon intensity indicators defined in section 4 should be applied.

3.2 The operational carbon intensity indicators defined in section 5 are encouraged to be additionally used by ships, where applicable, for trial purposes.

4 Operational carbon intensity indicator (CII) of individual ships for use in implementing regulation 28 of MARPOL Annex VI

In its most simple form, the attained annual operational CII of individual ships is calculated as the ratio of the total mass of CO 2 (M) emitted to the total transport work (W) undertaken in a given calendar year, as follows:

(1)

4.1 M ass of CO2 emissions (M)

The total mass of CO2 is the sum of CO2 emissions (in grams) from all the fuel oil consumed on board a ship in a given calendar year, as follows:

= × (2)

where:

---CHUNK---

SOURCE: MEPC.336(76).pdf
The total mass of CO2 is the sum of CO2 emissions (in grams) from all the fuel oil consumed on board a ship in a given calendar year, as follows:

= × (2)

where:

is the fuel oil type;
is the total mass (in grams) of consumed fuel oil of type in the calendar

year, as reported under IMO DCS; and
represents the fuel oil mass to CO 2 mass conversion factor for fuel oil type
, in line with those specified in the 2018 Guidelines on the method of
calculation of the attained Energy Efficiency Design Index (EEDI) for new ships
(resolution MEPC.308(73)), as may be further amended. In case the type of the
fuel oil is not covered by the guidelines, the conversion factor should be obtained
from the fuel oil supplier supported by documentary evidence.

4.2 Transport work (W)

In the absence of the data on actual transport work, the supply-based transport work (Ws) can
be taken as a proxy, which is defined as the product of a ship s capacity and the distance
travelled in a given calendar year, as follows:

Ws= C×Dt (3)

where:

C represents the ships capacity:

---CHUNK---

SOURCE: MEPC.336(76).pdf
be taken as a proxy, which is defined as the product of a ship s capacity and the distance
travelled in a given calendar year, as follows:

Ws= C×Dt (3)

where:

C represents the ships capacity:
- For bulk carriers, tankers, container ships, gas carriers, LNG carriers, ro- ro
cargo ships, general cargo ships, refrigerated cargo carrier and combination
carriers, deadweight tonnage (DWT)1 should be used as Capacity;
- For cruise passenger ships, ro- ro cargo ships (vehicle carriers) and ro- ro
passenger ships, gross tonnage (GT)2 should be used as Capacity; and
Dt represents the total distance travelled (in nautical miles), as reported under
IMO DCS.

1 Deadweight tonnage (DWT) means the difference in tonnes between the displacement of a ship in water of relative density of 1,025 kg/m3 at the summer load draught and the lightweight of the ship. The summer load draught should be taken as the maximum summer draught as certified in the stability booklet approved by the Administration or any organization recognized by it.

2 Gross tonnage (GT) should be calculated in accordance with the International Convention on Tonnage

---CHUNK---

SOURCE: MEPC.336(76).pdf
the Administration or any organization recognized by it.

2 Gross tonnage (GT) should be calculated in accordance with the International Convention on Tonnage Measurement of Ships, 1969.
/ shipattained CII M W=

j
jFC j
jFC
j

5 Operational carbon intensity indicator (CII) of individual ships for trial purpose

The following metrics are encouraged to be used for trial purposes, where applicable:

.1 Energy Efficiency Performance Indicator (EEPI)

= ×

.2 cbDIST

= ×

.3 clDIST

= ×

.4 EEOI, as defined in MEPC.1/Circ.684 on Guidelines for voluntary use of the ship energy efficiency operational indicator (EEOI).

In the formulas above:

the mass of CO2 (M), the ships capacity (C) and the total distance travelled (Dt) are identical with those used to calculate the attained CII of individual ships, as specified

---CHUNK---

SOURCE: MEPC.336(76).pdf
In the formulas above:

the mass of CO2 (M), the ships capacity (C) and the total distance travelled (Dt) are identical with those used to calculate the attained CII of individual ships, as specified
in section 4.1 and 4.2;

Dl means the laden distance travelled (in nautical miles) when the ship is loaded;

ALB means the number of available lower berths of a cruise passenger ship; and

Lanemeter means the length (in metres) of the lanes of a ro-ro ship.

***

RESOLUTION MEPC.336(76) (adopted on 17 June 2021)
2021 GUIDELINES ON OPERATIONAL CARBON INTENSITY INDICATORS AND THE CALCULATION METHODS

(CII GUIDELINES, G1)RESOLUTION MEPC.336(76) (adopted on 17 June 2021)
2021 GUIDELINES ON OPERATIONAL CARBON INTENSITY INDICATORS AND THE CALCULATION METHODS (CII GUIDELINES, G1)

---CHUNK---

SOURCE: MEPC.337(76).pdf
RESOLUTION MEPC.337(76) (adopted on 17 June 2021)
2021 GUIDELINES ON THE REFERENCE LINES FOR USE WITH OPERATIONAL CARBON INTENSITY INDICATORS (CII REFERENCE LINES GUIDELINES, G2)MEPC 76/15/Add.2
Annex 11, page 1

I:\MEPC\76\MEPC 76-15-Add.2.docx
ANNEX 11

RESOLUTION MEPC.337(76)
(adopted on 17 June 2021)

2021 GUIDELINES ON THE REFERENCE LINES FOR USE WITH OPERATIONAL CARBON INTENSITY INDICATORS (CII REFERENCE LINES GUIDELINES, G2)

THE MARINE ENVIRONMENT PROTECTION COMMITTEE,

RECALLING Article 38(a) of the Convention on the International Maritime Organization concerning the functions of the Marine Environment Protection Committee conferred upon it by international conventions for the prevention and control of marine pollution from ships,